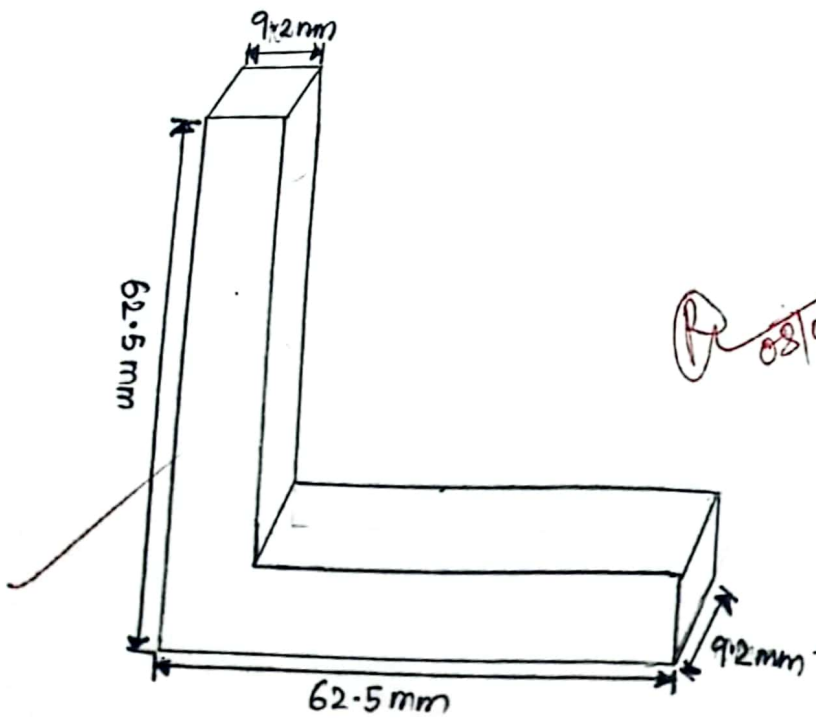


MILD STEEL ROD



P 08/02/2023



HARCOURT BUTLER TECHNOLOGICAL UNIVERSITY

KANPUR

WORKSHOP PRACTICE I

FULL NAME..Muskan Sharma..... CLASS..1st ME.....

SHOP..Black Smithy Shop..... DATE..20/12/2022.....

WORKSHOP PRACTICE-01

Objective: Study of tools and equipments to make a 'L' shaped job by hand forging operation as per given drawing diagrams.

Date of Starting: 20/12/2022

Date of Finishing: 20/12/2022

Materials Required:

Mild Steel Rod (125mm X 10 mm)

Steam Coal

Tools and Equipments used:

1) Measuring and marking tools:

i) Smith Square

ii) Template

2) Holding and Supporting tools:

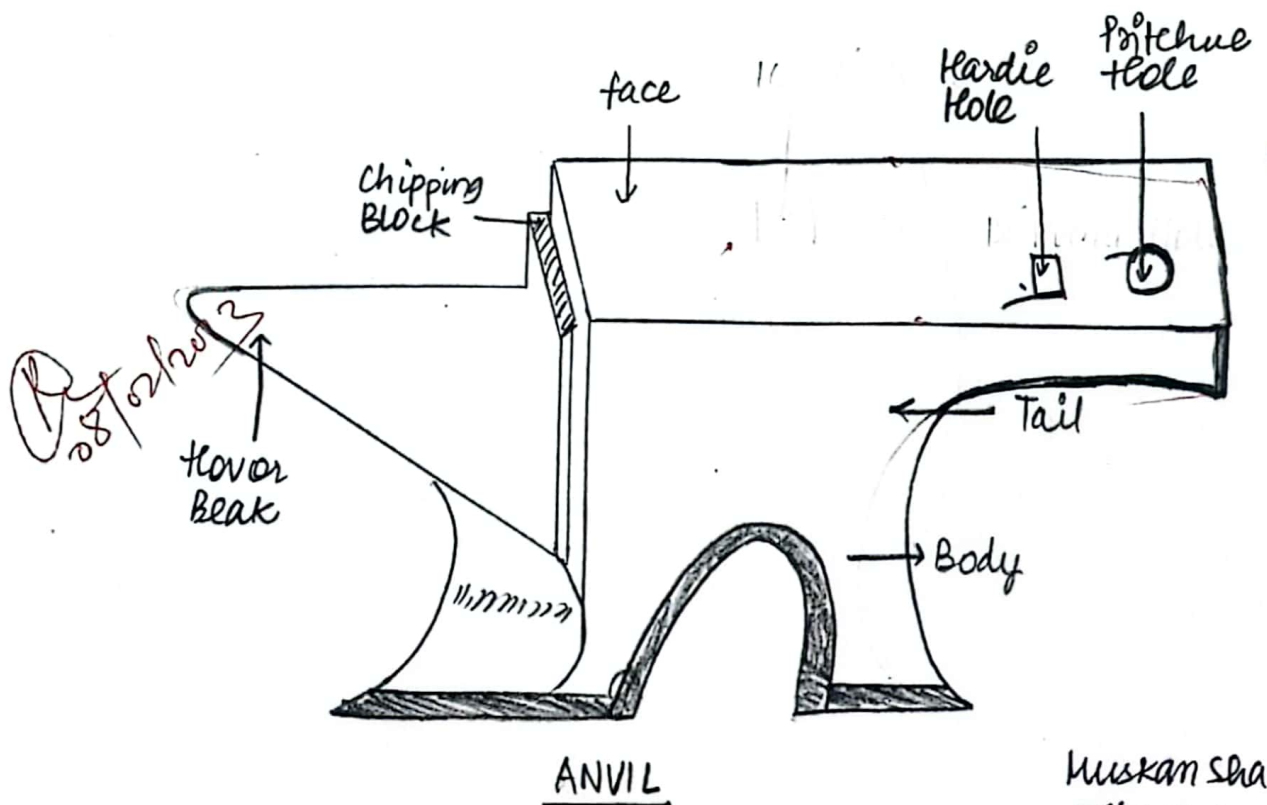
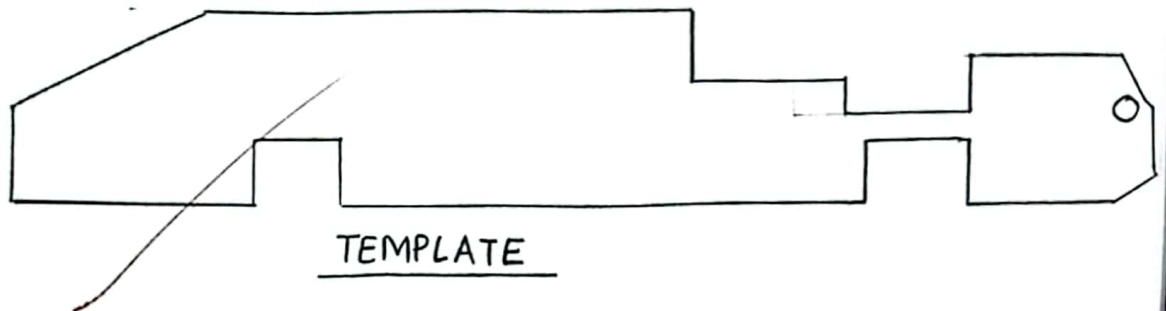
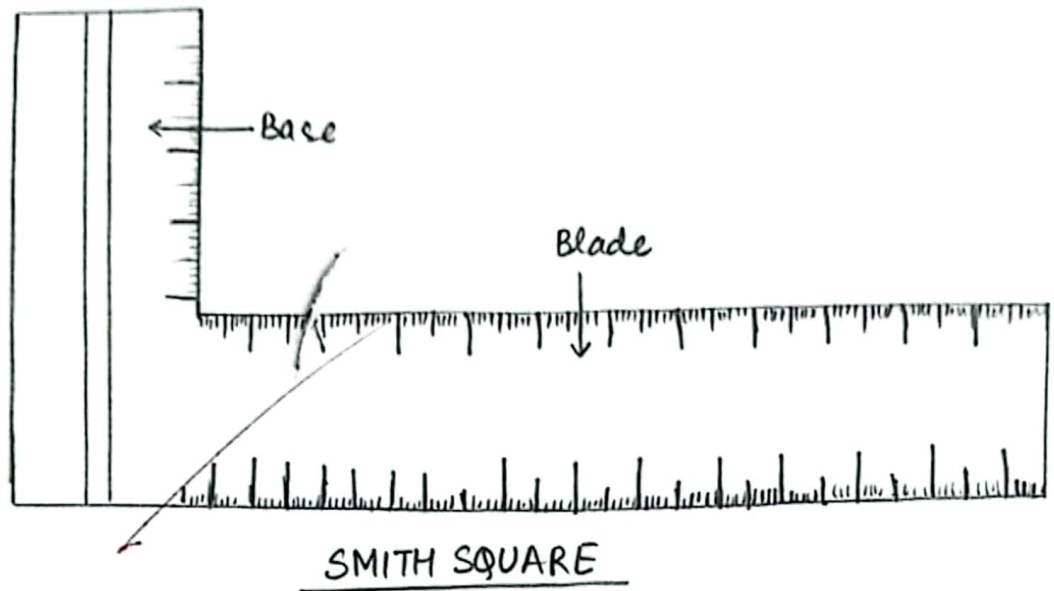
i) Anvil

ii) Swage block

iii) Tongs

iv) Leg Vice

Paras



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3) Striking Tools:

i) Hammer

4) Cutting Tools:

i) Chisel

a) Hot Chisel

b) Cold Chisel

ii) Hardie

5) Miscellaneous Tools:

i) Swage

ii) Fullers

iii) Punch and drift

iv) Open Hard Furnace

Description of Tools and Equipments:

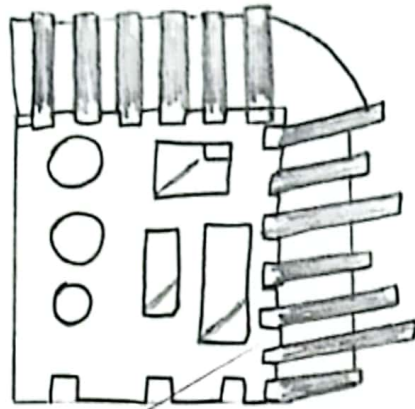
1) Marking and Measuring Tools:

a) Smith Square: Smith Square is used for measuring, marking and checking the accuracy of Right Angled Rod. It is available in two sizes 9" x 8" and 12" x 24".

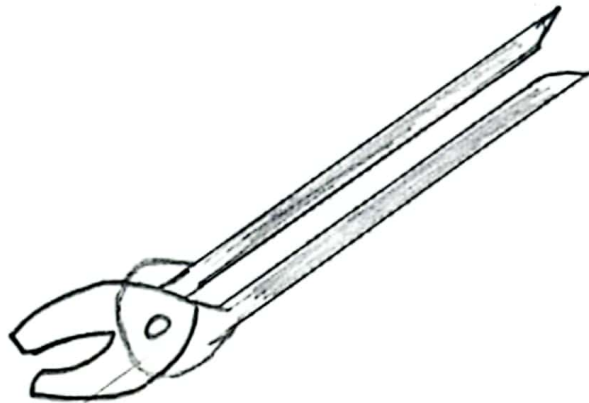
b) Template: Template are locally manufactured tool. Dimensions are cut appropriate to measure hot jobs.

2) Holding and Supporting Tools:

light



SWAGE BLOCK

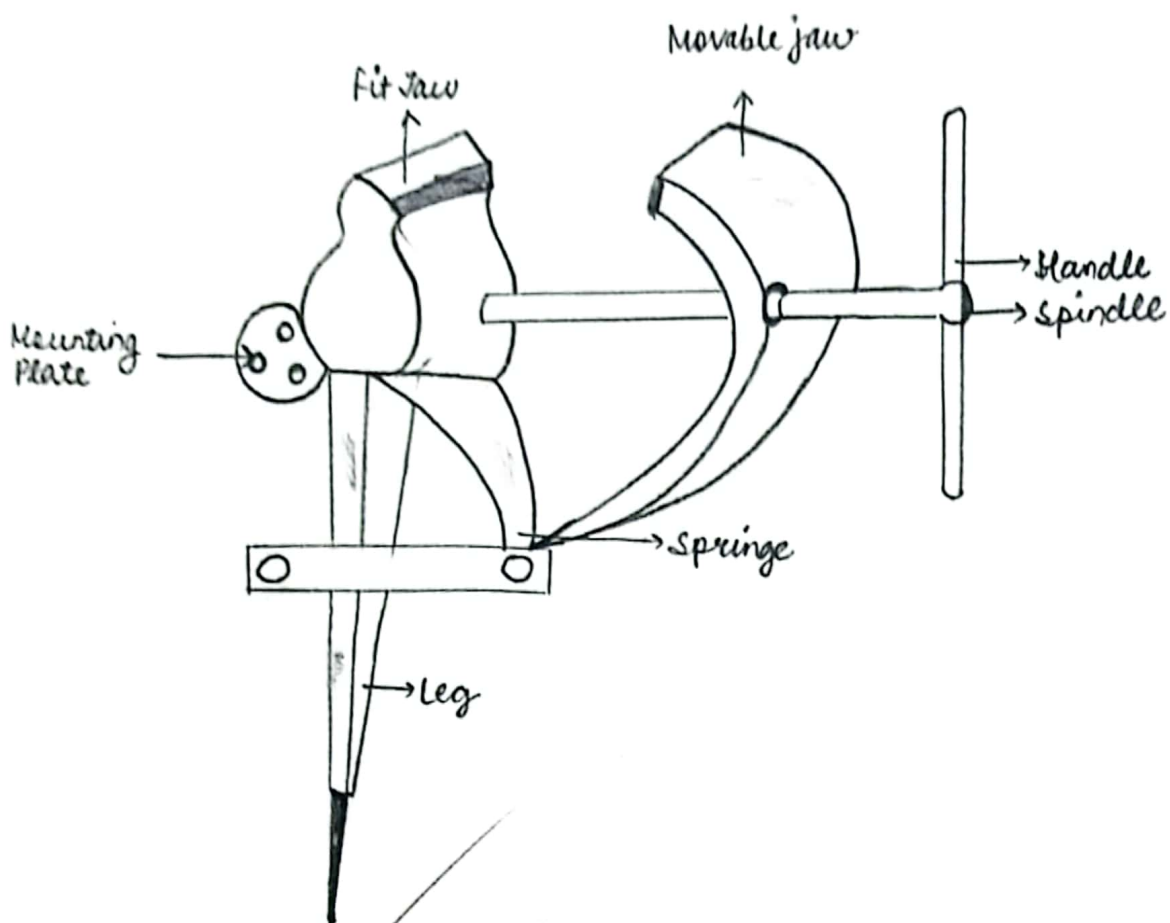


OPEN MOUTH TONG

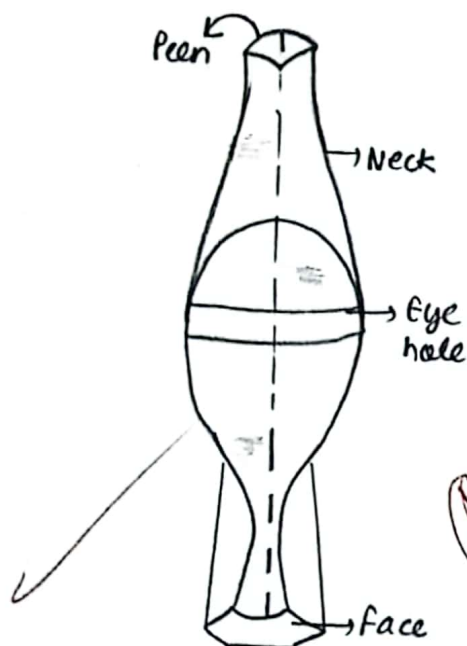
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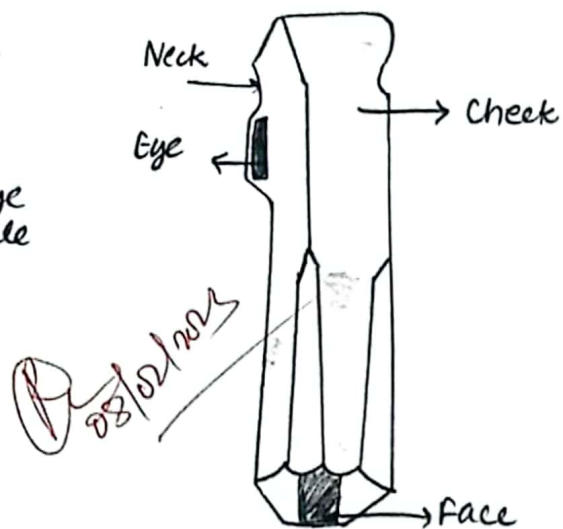
- a) Anvil: It provides necessary support during forging by resisting blows rendered to the job. It performs operations such as bending, shaping etc. Its body is generally made of cast iron, wrought iron, or mild steel with a hardened layer of about 20 to 25 mm thick. The ham or beak is used for bending or forming required shapes. The hardie hole which is square in shape is used to hold square shape tools like hardie, swages, and fullers. The pritching hole is used for bending small rods and punching holes in the sheet. Anvil generally weigh from 25 to 250 kg.
- b) Swage Block: The swage block is generally made of cast iron and has round, square, rectangular and half round grooves. In addition to this, it is provided with holes which are useful for holding jobs while bending and breaking up heads. The swage block is generally supported supported at a suitable height on a stand which is adaptable to hold it flatwise and on its edge.
- c) Tongs: Tongs are holding tools made of mild steel, normally used for gripping hot metals. Tongs come in a great variety of shapes and sizes. Most commonly used are the closed mouth tong, Open mouth tong and pickup tong.
- d) Leg Vice: It is used to hold hot jobs for further operations like hammering and bending etc. This is generally made up of mild steel. It is specified by length of its jaws.



LEG VICE

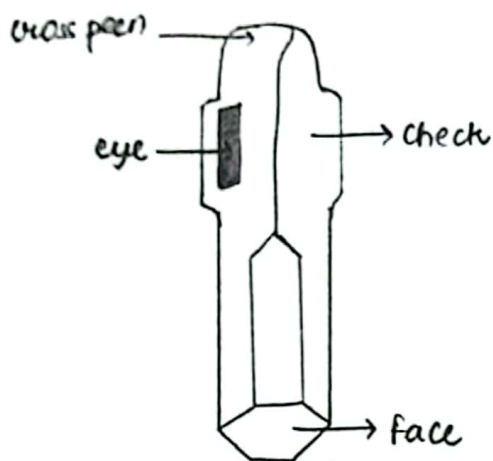


BALL PEEN HAMMER

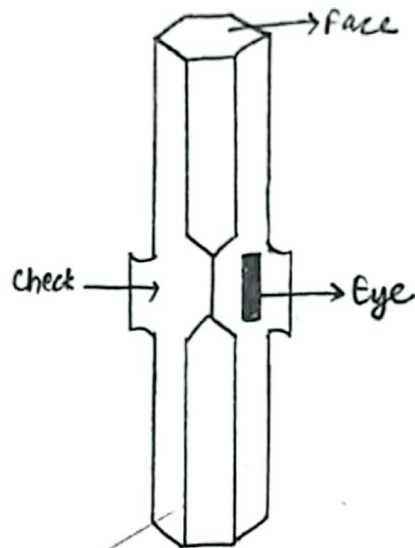


STRAIGHT PEEN SLEDGE HAMMER

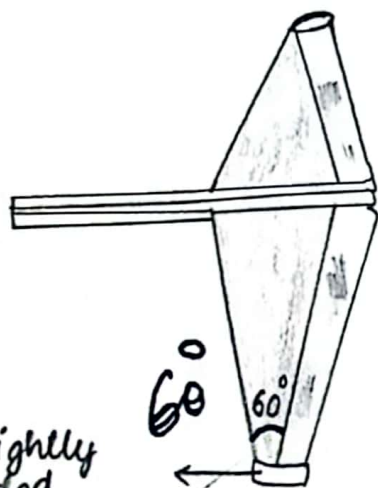
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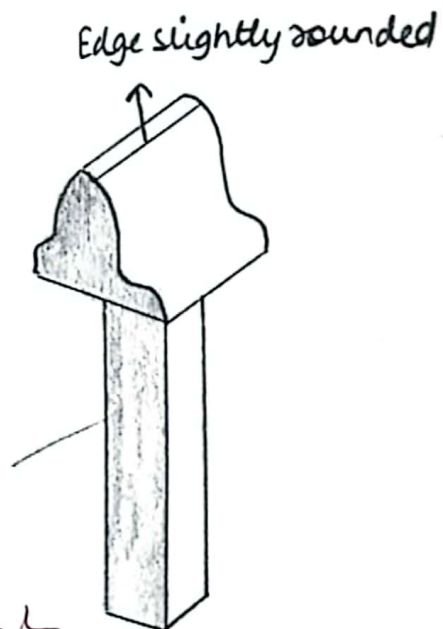
CROSS PEEN HAMMER



DOUBLE PEEN HAMMER



COLD CHISEL



Mallet

Post 20/02/2025

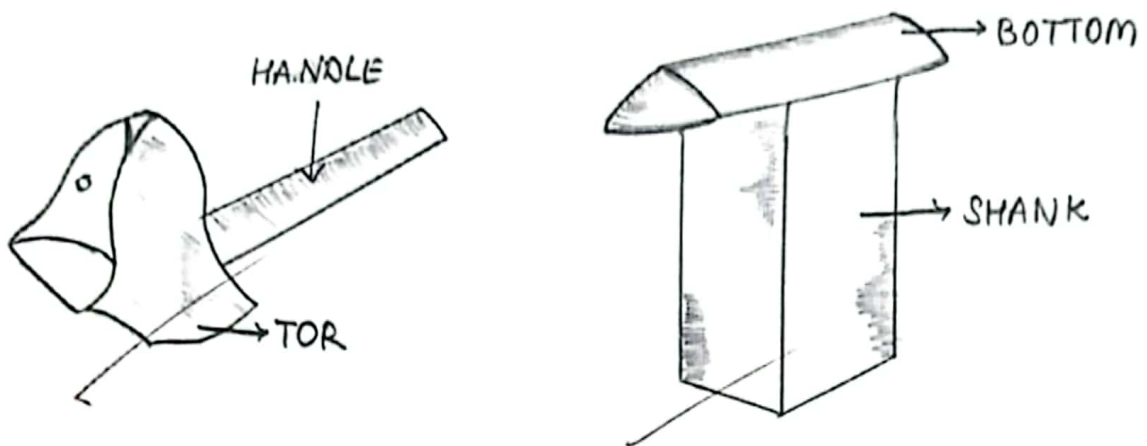
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3) Striking Tools:

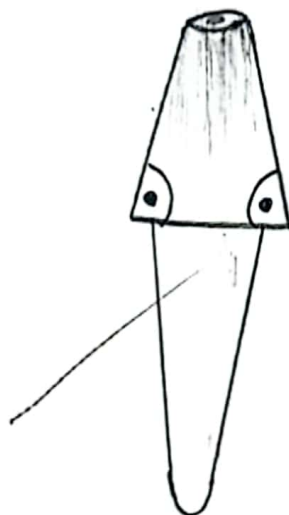
- a) Ball Peen Hammer: It is the lighter hammer used by the blacksmith by one hand during the cast steel operation. used in fitting shops.
- b) Cross Peen Hammer: In this hammer, the pin of its tip is perpendicular to axis of its edge. It is also a lighter hammer and used by one hand by Blacksmith.
- c) Straight Peen Hammer: It is similar to Cross-Peen Hammer except that pin is parallel to the axis of its edge.
- d) Sledge Hammer: It is used for heavy blows by the smith for heavy reduction in metal size and many other operations.

4) Cutting Tools:

- a) Chisels: Chisels are used for cutting metals and for breaking. They may be cold or hot depending upon whether the metal to be cut is cold or hot. The difference between the two chisels is in the edges. The types of chisels are :-
 - i) Hot Chisel: The hot chisel are made up of low carbon steel and has the cutting angle of 30° . The hardening of the edge is not necessary as the hot metal would soften it. These types of chisel is fitted with wooden shaft.
 - ii) Cold Chisel: The chisels are made of high carbon steel with a cutting angle of about 60° & its edge is hardened and tempered.



A PAIR OF FULLERS



A PUNCH

Posters

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b) Hardy: The hardy are generally used in pairs. The bottom chisel has square shank and fits in the square hardy held in the anvil face.

5) Miscellaneous Tools:

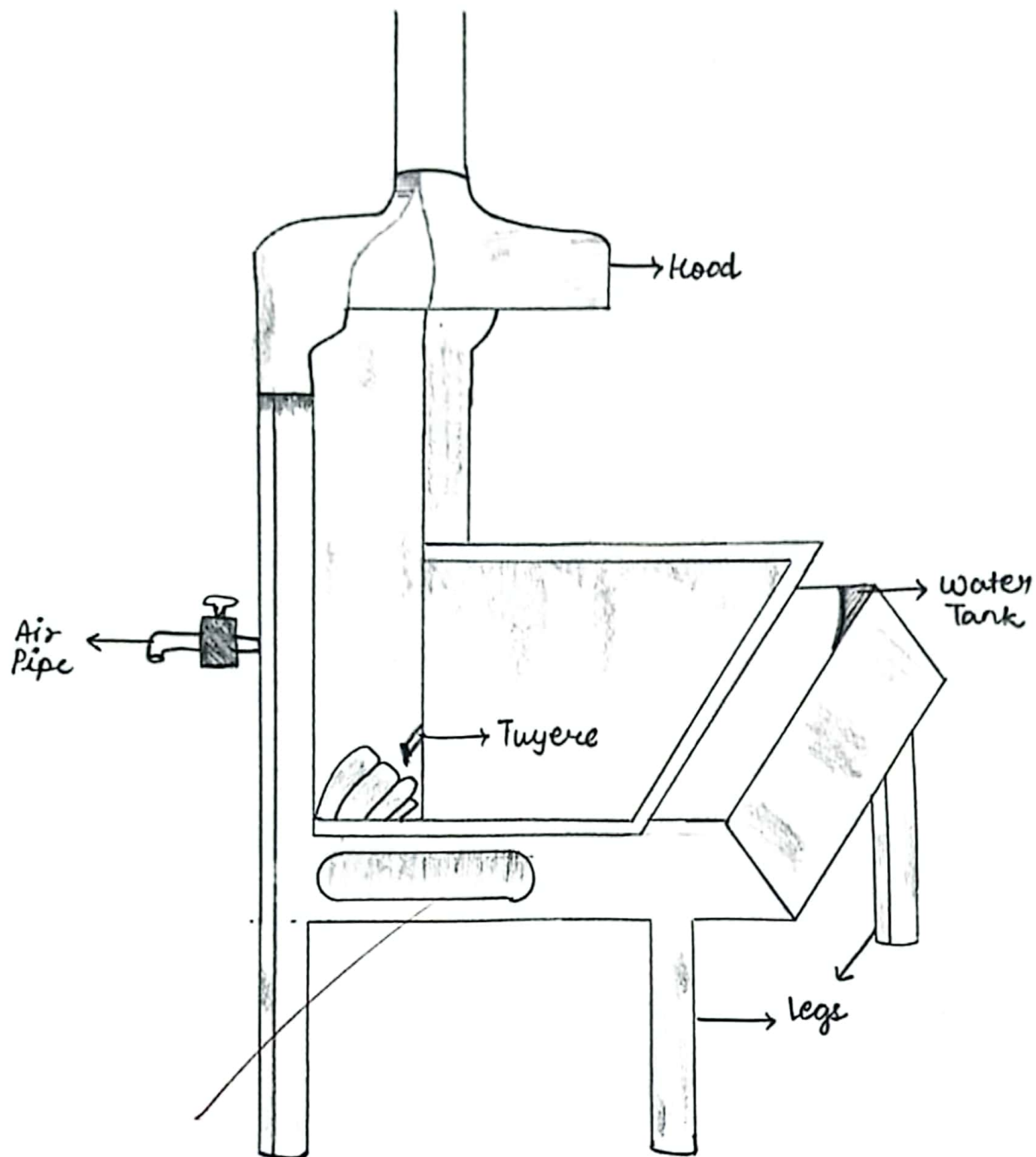
a) Swages: These are made of high carbon steel in two parts called top and bottom edges. They are used for increasing the length of circular rod or finishing the circular surface of a job after forging.

b) Fullers: These tools are made of high carbon steel in different sizes to suit the various types of jobs. Their working edges are normally rounded.

c) Punches and drift: Blacksmith's punches are made of tool steel and are hardened and tempered. Punches are usually made in three forms: Square, Round and Oval. Drifts are used for enlarging punched holes.

Procedure:

- 1) Job is heated in the furnace to the cherry red colour.
- 2) Hold the job by flat bit tong and place the job on the hard face of the anvil.
- 3) Upsetting is done by ball peen hammer throughout the length of the Round bar.
- 4) Keep upside down and forge the bottom portion of the bar.
- 5) At this stage ovality is existing on the sides. Now ovaled



08/04/2023 SMITH'S FORGE

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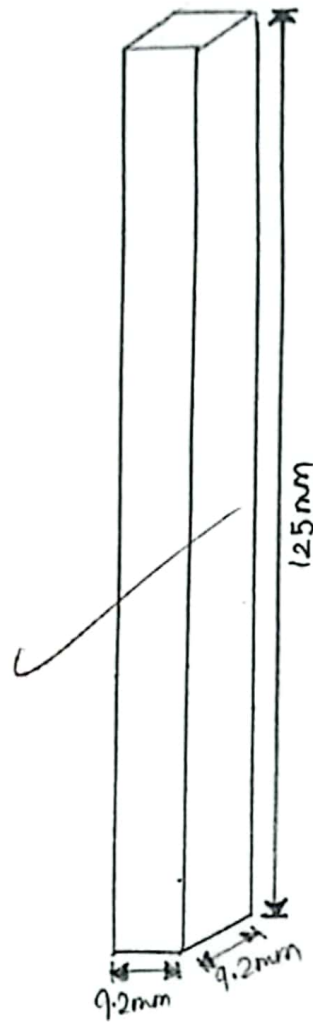
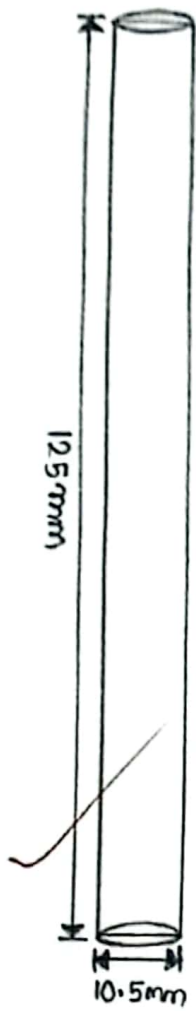
portion is drawn out to make the square cross section.

- 6) Job is again heated and centre upsetting is done.
- 7) Heated job is placed on the face of the Anvil, bent at right angle from the centre.
- 8) Twisting and buckling is removed by hammering on the top and bottom face of the job.
- 9) Finishing is done by heating the job at the dull red colour and allowed to cool down in air gradually.

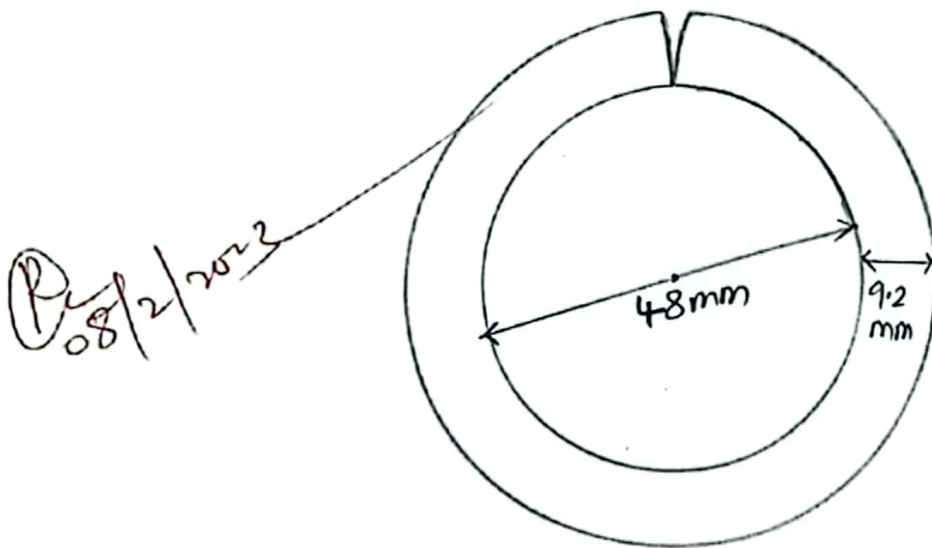
Precautions:

- 1) Always use proper clothing and shoes.
- 2) Use suitable tongs for good gripping.
- 3) Avoid over heating and under heating of workpiece.
- 4) Avoid touching hot jobs.

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08/02/2023



MILD STEEL ROD



RING

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HARCOURT BUTLER TECHNOLOGICAL UNIVERSITY

KANPUR

WORKSHOP PRACTICE

FULL NAME... Muskan Sharma..... CLASS... 1st ME.....

SHOP... Black Smithy Shop..... DATE... 03/01/2023.....

WORKSHOP PRACTICE - 02

Objective: Study of Tools and Equipments to make a ring from the given piece of mild steel rod by hand forging operation as per given diagram.

Date of Starting: 03/01/2023.

Date of Finishing: 03/01/2023

Materials Required:

Mild steel rod (²⁵~~15~~mm x 10mm)

Steam coal

Tools and Equipments Required:

1) Marking and Measuring Tools:

i) Smith square

ii) Template

2) Holding and Supporting Tools:

i) Anvil

ii) Surge Block

iii) Tong

iv) Leg vice

Paras

3) Striking Tools:

- i) Ball Peen Hammer
- ii) Cross Peen Hammer
- iii) Straight Peen Hammer
- iv) Sledge Hammer

4) Cutting Tools:

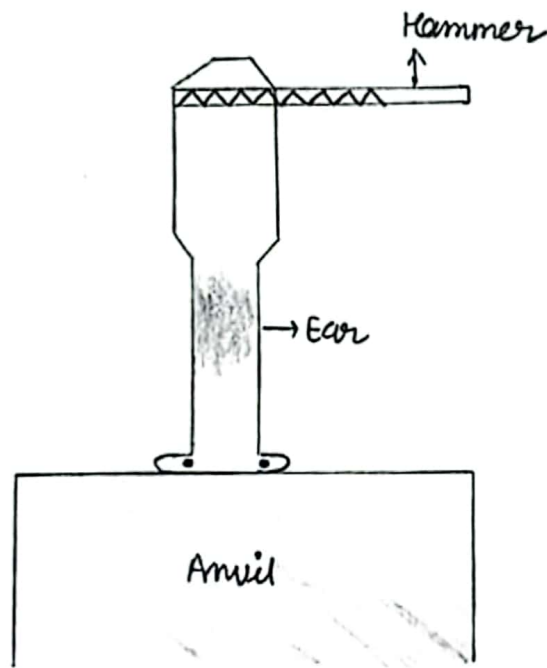
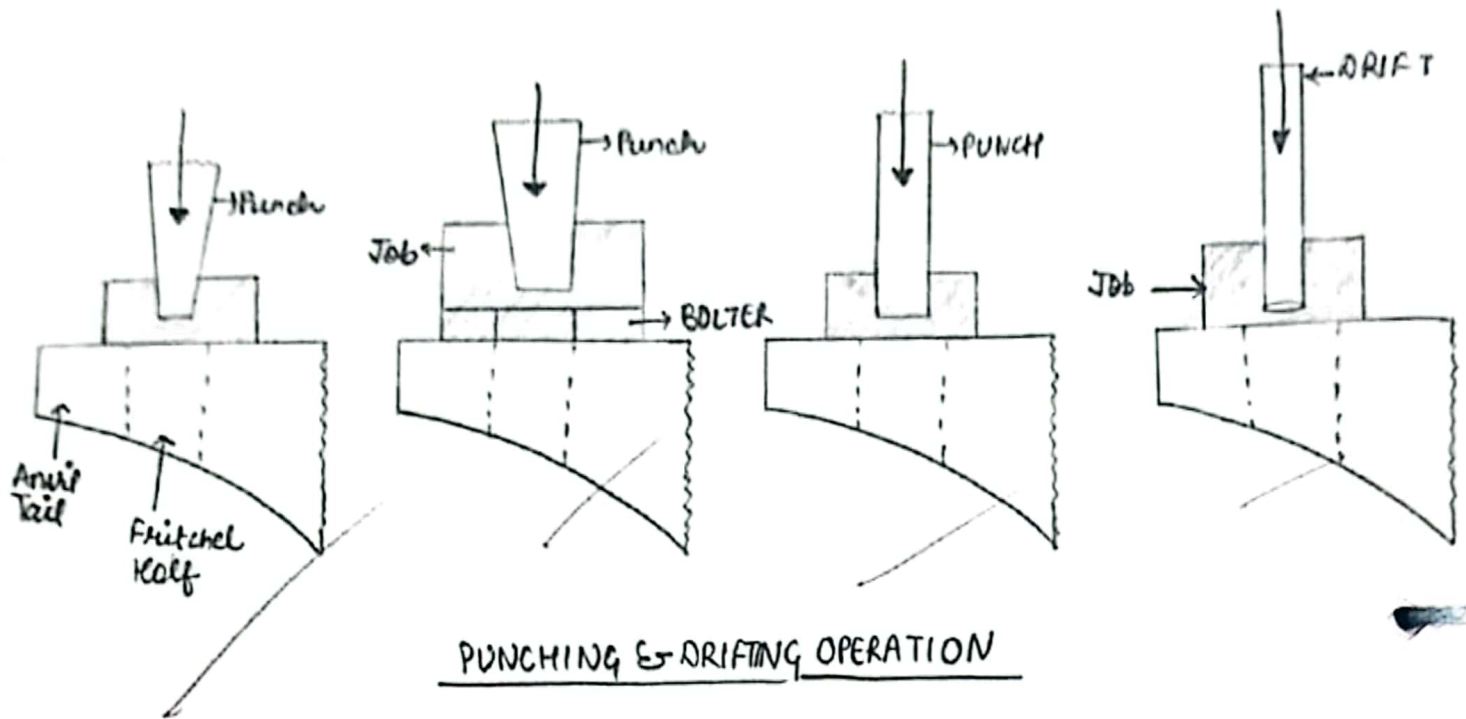
- i) chisels
 - a) Hot chisel
 - b) Cold chisel
- ii) Hardy

5) Miscellaneous Tools:

- i) Swage
- ii) Punch and drift
- iii) Fuller
- iv) Smith's forge and hearth

Brief description of forging process done in Blacksmith work:

- 1) Upsetting and Tumbling: It is the process through which the cross-section of a metal piece is increased with a corresponding reduction in length.
- 2) Drawing down or drawing out: It is exactly the reverse process to that of upsetting in the sense that, it is employed. When reduction in thickness, width or both of a bar is desired with a corresponding increase in its length.



Prashant

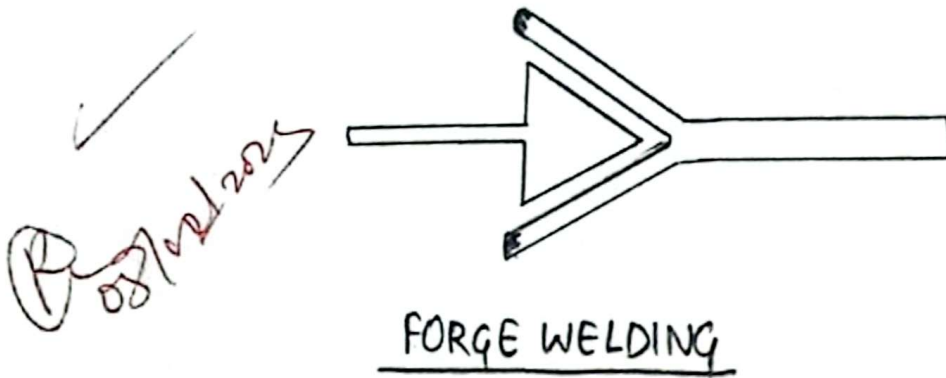
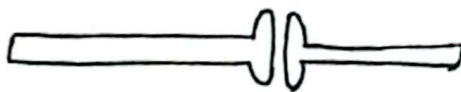
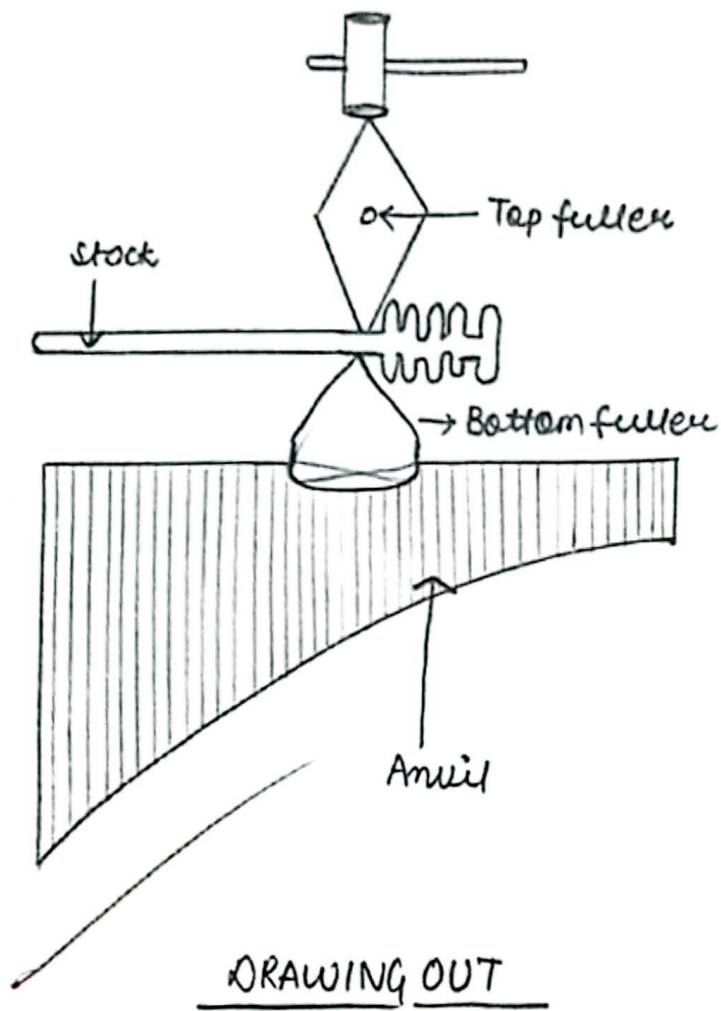
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- 3) Cutting: Cutting of a metal in hot or cold state is done by measure of hot or cold chisels respectively. It is required in removing extra metal from job before finishing it.
- 4) Bending: This can be done to produce different types of bent shapes such as angles, ovals and ~~ovals~~ circles. Any desired angle or curvature can be made through this operation.
- 5) Punching: This term refers to the operation in which a punch is forced through a job to produce a job. Punching is usually followed by drifting. In this, a tool known as drift is made to pass through the punched hole to produce a finished hole of the required size.
- 6) Setting down: It is the operation through which the rounding of the corner is removed to make it square by means of a set hammer.
- 7) Welding: In this operation, two pieces of same metal which are to be joined together are heated in the hearth to the proper temperature when they have acquired the welding heat they are withdrawn from the furnace and joined together by the application of external pressure.

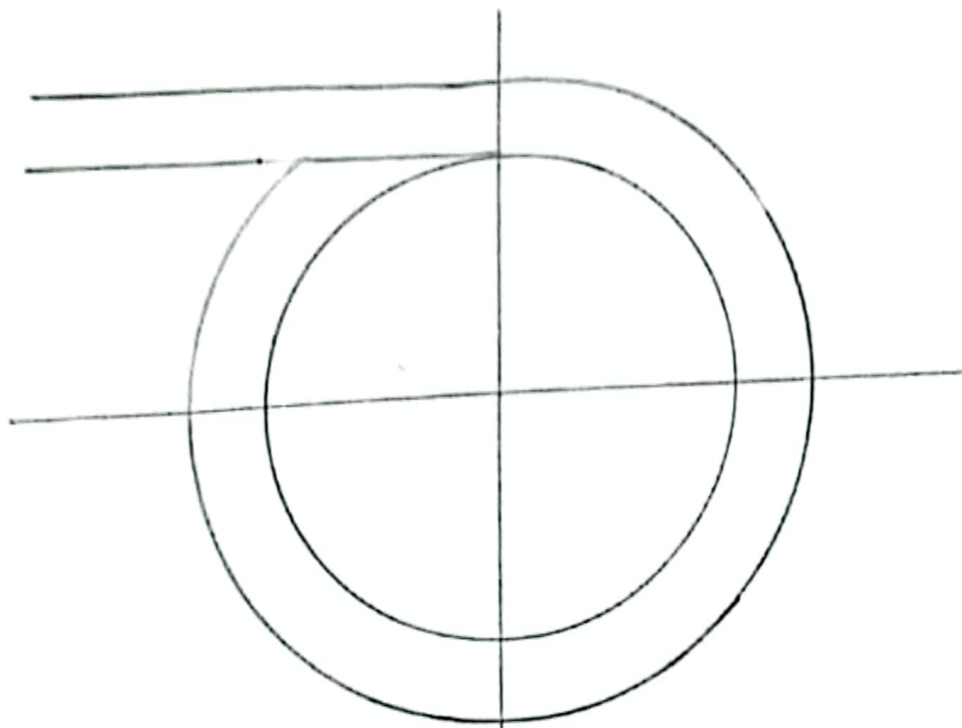
Procedure:

- 1) Job is heated in the open hearth furnace to the cherry red colour.
- 2) Heat the job by flat bit tong and place the job on the hard face

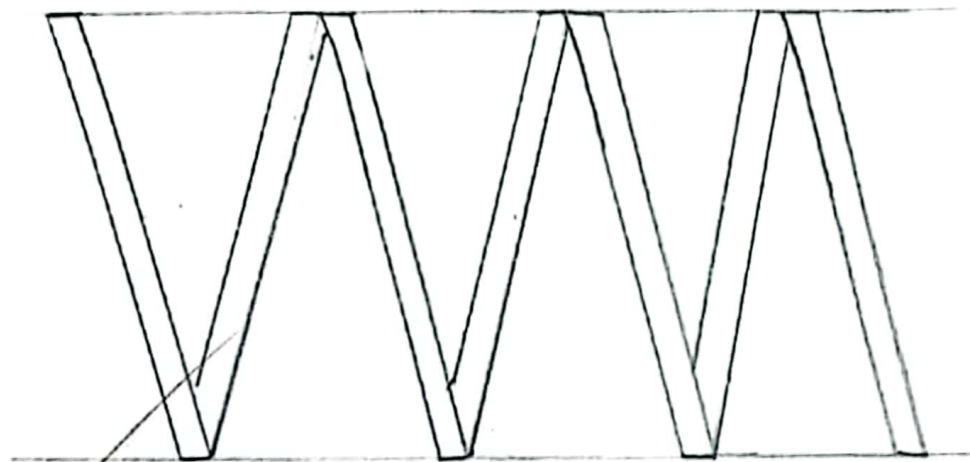
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BENDING AROUND BAR TO FORM HELICAL
SPRING



FINISHING JOB AFTER BENDING

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of the Anvil.

- 3) Upsetting is done by Ball Peen hammer throughout the length of the round bar.
- 4) Keep upside down and forge the bottom portion of the bar.
- 5) Oval portion of sides are drawn out to make the square cross section.
- 6) Square cross-section rod is heated to cherry red colour and placed on the horn of the Anvil.
- 7) Pressure is applied by the hammer to form a Circular ring.
- 8) Twisting and Buckling is removed by hammering on the top and bottom face of the jobs.
- 9) Finishing is done by heating the job at the dull red colour and allowed to cool down in the air gradually.

Precautions:

- 1) Proper clothing and shoes should be used.
- 2) Damaged and wrongly positioned tools should be avoided.
- 3) Job should be heated to proper forging limits.
- 4) Do not stand near the furnace.
- 5) Job should be held by the tongs properly.

(P) 03/01/2023